

3.6 Libraries in Python

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے import سے پروگرام میں لایا جاتا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 21 of 24

3.6 Libraries in Python

LEARN IT

A **library** (also called a **module**) is a collection of pre-written code that offers ready-made functions. Instead of writing a square-root algorithm from scratch, you import `math` and call `math.sqrt()`. Libraries save time, reduce bugs and let you stand on other programmers' work.

There are five common ways to import a library. Method 3 (import only what you need) is the best habit: it is clearer and uses less memory than `from math import *`.

Method	Syntax	How you call the function
1. Entire module	<code>import math</code>	<code>math.sqrt(25)</code>
2. Alias (nickname)	<code>import math as m</code>	<code>m.sqrt(16)</code>
3. Specific function	<code>from math import sqrt</code>	<code>sqrt(36)</code>
4. Several functions	<code>from math import sqrt, pi</code>	<code>sqrt(9), pi</code>
5. Everything (*)	<code>from math import *</code>	<code>sqrt(64)</code> — avoid this

WORKED EXAMPLE

Five imports, one result each:

```
import math
print(math.sqrt(25))
import math as m
print(m.sqrt(16))
from math import sqrt
print(sqrt(36))
from math import sqrt, pi
print(pi)
from math import *
print(sqrt(64))
```

OUTPUT

```
5.0
4.0
6.0
3.141592653589793
8.0
```

COMMON MISTAKE

`from math import *` dumps every name into your program. If you later write a function called `sqrt`, you silently overwrite the library one. Prefer methods 1–4.

اردو میں سمجھیں

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے `import` سے پروگرام میں لایا جاتا ہے۔ سب سے اچھا طریقہ یہ ہے کہ صرف وہ فنکشن درآمد کریں جس کی ضرورت ہو، ستارہ `*` سے سب کچھ نہ کھینچیں۔

CHECK YOURSELF

1. Write the line that lets you call `ceil` without writing `math.` in front.
2. Why is an alias such as `import pandas as pd` useful?

Answers: 1. `from math import ceil`. 2. The library name is long; a short nickname keeps later lines short and is a community convention.

Next lecture: 3.6.1 Built-in Libraries